The Limits of Utility Maximization under Stable Preferences: Evidence for Complementarity and Context Dependence

Evidence-Based Framework Research Team*

January 2026

Abstract

We examine the empirical and theoretical limits of utility maximization under stable preferences. Drawing on evidence from six domains—social preferences, identity and self-image, reference dependence, framing effects, preference construction, and temporal discounting—we demonstrate that the stability assumption is not merely a simplification but a substantive restriction that excludes central economic phenomena. The deeper problem is *structural*: utilities exhibit interdependence across agents ($\partial^2 U_i/\partial x_i \partial x_j \neq 0$), time ($\partial U_t/\partial a_{t-1} \neq 0$), and context ($\partial U/\partial \Psi \neq 0$). These cross-derivatives are precisely the *complementarities* that generate coordination problems, multiple equilibria, and path dependence. We formalize these interdependencies within a unified framework where complementarity C and context Ψ are explicit parameters. The framework does not reject utility maximization but embeds it within a richer

*FehrAdvice & Partners AG. Correspondence: research@fehradvice.com. We thank the behavioral economics community for decades of rigorous experimental evidence that makes this synthesis possible.

structure that accommodates 50+ years of behavioral evidence. We derive testable predictions about when behavioral deviations are large versus small, and propose experimental designs to estimate context-dependent parameters.

JEL Codes: D01, D91, D87, C92

Keywords: Utility maximization, stable preferences, social preferences, reference dependence, complementarity, context dependence

1 Introduction

The classical framework in economics assumes that agents maximize well-defined utility functions with stable preferences. This assumption is not arbitrary—it serves crucial theoretical functions: tractability (fixed preferences enable optimization), prediction (stable preferences allow demand estimation), welfare analysis (unchanging preferences define a welfare criterion), and identification (without stability, any behavior can be rationalized as "rational").

Yet five decades of experimental and field evidence have accumulated demonstrating systematic violations of this assumption. The question is no longer *whether* violations occur but *what structure* underlies them. This paper argues that the violations share a common feature: **utility interdependence**. Specifically:

- **Cross-agent interdependence:** $\frac{\partial U_i}{\partial x_j} \neq 0$ (social preferences)
- **Temporal interdependence:** $\frac{\partial U_t}{\partial a_{t-1}} \neq 0$ (habits, identity)
- **Intertemporal inconsistency:** $U_t(c_t, c_{t+1}) \neq \delta \cdot U_{t+1}(c_{t+1}, c_{t+2})$ (present bias)
- **Context interdependence:** $\frac{\partial U}{\partial \Psi} \neq 0$ (framing, defaults)

These interdependencies generate non-zero second derivatives—*complementarities*—that are precisely what classical theory assumes away.

1.1 The Stigler-Becker Defense and Its Limits

Stigler & Becker (1977) argued that apparent preference changes can always be reinterpreted as responses to changing constraints: "De gustibus non est disputandum"—tastes don't change; only opportunities do. For example, addiction can be modeled as a stable preference over consumption and a stock variable $S_t = \sum_{\tau<t} c_\tau$:

$$U(c_t, S_t) \quad \text{where } S_t \text{ is consumption stock} \tag{1}$$

This reinterpretation saves stability but at a cost: it makes the theory unfalsifiable. Any behavior pattern can be rationalized with a sufficiently complex constraint structure. The question becomes empirical: do we gain predictive power from treating preferences as stable, or from explicitly modeling their dependence on social context, reference points, and time?

1.2 Contribution

We make three contributions. First, we synthesize evidence from six domains where the stable preferences assumption fails, showing that these are not isolated anomalies but manifestations of a common structure. Second, we formalize the underlying interdependencies within a framework where complementarity C and context Ψ are explicit parameters. Third, we derive testable predictions about when behavioral deviations are large versus small, enabling quantitative prediction rather than mere cataloguing of anomalies.

2 Evidence: Six Domains of Instability

2.1 Domain 1: Social Preferences

The Evidence. Thousands of experiments demonstrate that people care about others' outcomes:

- **Ultimatum games:** Responders reject unfair offers, sacrificing money (Güth et al., 1982)
- **Dictator games:** Dictators share even when anonymous (Forsythe et al., 1994)
- **Public goods games:** Many contribute despite dominant strategy to free-ride (Andreoni, 1988)
- **Trust games:** Investors trust and trustees reciprocate (Berg et al., 1995)

The ultimatum game illustrates most starkly: a proposer divides \$10, and a responder accepts or rejects (rejection yields zero for both). Classical prediction: offer \$0.01, accept. Actual behavior: modal offers are \$4–5; offers below \$2 are frequently rejected.

Proposition 1 (Social Preference Implication)**.** *Utility is not $U_i(x_i)$ but $U_i(x_i, x_j)$. The cross-derivative is non-zero:*

$$\frac{\partial^2 U_i}{\partial x_i \partial x_j} \neq 0 \tag{2}$$

This is precisely the complementarity that classical theory assumes away.

Heterogeneity. The distribution of types is well-established (Fehr & Charness, 2024):

Type	**Students**	**General Population**
Selfish	40–60%	20–30%
Inequality averse	10–15%	15–20%
Altruistic	25–45%	45–55%

Table 1: Distribution of social preference types

Critically, the *majority* of individuals exhibit social preferences; purely selfish individuals are a minority, especially in representative samples.

Context Dependence. Social preferences vary with context:

- Higher in face-to-face interactions
- Higher with in-group members
- Lower in competitive frames
- Sensitive to merit: inequality from effort is more accepted than inequality from luck (Cappelen et al., 2007; Almas et al., 2020)

Thus the complementarity itself is context-dependent: $C_{ij} = C_{ij}(\Psi)$.

2.2 Domain 2: Identity and Self-Image

The Evidence. People sacrifice material payoffs to maintain consistent self-image. The "moral wiggle room" experiment (Dana et al., 2007) is instructive: subjects chose between (A) \$6 for self, \$1 for other, or (B) \$5 for self, \$5 for other. Many chose B (fair). But when given the option to "not know" what the other receives, many chose ignorance and then A. They preferred not knowing they were being unfair.

Proposition 2 (Identity Utility)**.** *Utility includes self-image:*

$$U = U^{material} + U^{IDN} \tag{3}$$

where U^{IDN} is identity utility—the value of being (or seeing oneself as) a good person.

Identity creates intertemporal linkage:

$$\frac{\partial^2 U}{\partial a_t \partial a_{t+1}} > 0 \tag{4}$$

Behaving as type X today makes it easier to see oneself as type X tomorrow. This mechanism underlies habit, character, and culture.

2.3 Domain 3: Reference Dependence and Loss Aversion

The Evidence. Kahneman & Tversky (1979) demonstrated that outcomes are evaluated relative to a reference point, with losses weighted more than gains:

$$v(x) = \begin{cases} x^{\alpha} & \text{if } x \geq 0 \\ -\lambda(-x)^{\beta} & \text{if } x < 0 \end{cases} \tag{5}$$

with $\lambda \approx 2$ (loss aversion coefficient).

The endowment effect (Knetsch et al., 1990) provides clean evidence: subjects given mugs

or chocolate bars trade far below 50%—they value what they have more than equivalent items they don't have.

Proposition 3 (Reference Point Dependence)**.** *The reference point r is part of context:*

$$U = U(x - r(\Psi)) \tag{6}$$

where $r(\Psi)$ depends on status quo, expectations (Kőszegi & Rabin, 2006), social comparison, and aspirations.

2.4 Domain 4: Framing and Context Effects

The Evidence. Identical choice problems elicit different responses depending on framing:

- **Asian Disease Problem:** "200 saved" vs. "400 die" reverses preferences (Tversky & Kahneman, 1981)
- **Default effects:** Germany (opt-in): 12% organ donation; Austria (opt-out): 99% (Johnson et al., 2003)
- **Anchoring:** Arbitrary numbers influence willingness to pay

Proposition 4 (Frame Dependence)**.** *Choices depend on presentation:*

$$\textit{Choice}(A, B; \Psi_1) \neq \textit{Choice}(A, B; \Psi_2) \tag{7}$$

even when A and B are objectively identical.

This directly violates stable preferences, which requires: same options $\Rightarrow$ same choice.

2.5 Domain 5: Preference Construction

The Evidence. For many choices, preferences are not "retrieved" but "constructed" at decision time. Contingent valuation studies show willingness to pay varies wildly with question

order, starting points, and comparison goods.

Proposition 5 (Constructed Preferences)**.** *For complex goods, preferences emerge from deliberation in context:*

$$U(x; \Psi) = f(\textit{construction process}(\Psi)) \tag{8}$$

This is especially true for emotional, social, identity, and ecological utility.

2.6 Domain 6: Temporal Preferences and Present Bias

The Evidence. Laibson (1997) formalized quasi-hyperbolic discounting:

$$U_t = u(c_t) + \beta \sum_{\tau=1}^{T} \delta^\tau u(c_{t+\tau}) \quad \text{where } \beta < 1 \tag{9}$$

Empirical estimates consistently find $\beta \approx 0.7$ and $\delta \approx 0.99$ (Frederick et al., 2002; Cohen et al., 2020). This implies:

- Substantial present bias (30% premium on immediate rewards)
- Time-inconsistent behavior: plans made today are not followed tomorrow
- Demand for commitment devices (Ashraf et al., 2006; Bryan et al., 2010)

Proposition 6 (Temporal Context)**.** *Temporal preferences constitute a Ψ_T dimension of context:*

$$\Psi_T = (\beta, \delta, \hat{\beta}) \tag{10}$$

where $\hat{\beta}$ reflects sophistication about own bias. This interacts with other context dimensions:

- $\Psi_T \times \Psi_C$*: Cognitive load worsens self-control*
- $\Psi_T \times \Psi_I$*: Institutions can provide commitment*

3 The Deeper Problem: Structural Interdependence

Each domain above could be treated as a "behavioral anomaly"—a deviation requiring a patch. But the deeper issue is *structural*: utilities are interdependent in ways that classical theory cannot accommodate.

Definition 1 (Four Types of Interdependence). *1.* ***Cross-agent:*** $\frac{\partial U_i}{\partial x_j} \neq 0$ *(my utility depends on your outcome)*

2. ***Temporal:*** $\frac{\partial U_t}{\partial a_{t-1}} \neq 0$ *(current utility depends on past actions)*

3. ***Intertemporal inconsistency:*** $U_t(\cdot) \neq \delta \cdot U_{t+1}(\cdot)$ *(time preferences not constant)*

4. ***Context:*** $\frac{\partial U}{\partial \Psi} \neq 0$ *(utility depends on institutional environment)*

Proposition 7 (Complementarity Implication). *All four types generate non-zero second derivatives:*

$$C_{ij} = \frac{\partial^2 U_i}{\partial x_i \partial x_j} \neq 0 \tag{11}$$

This is the structural feature that creates coordination problems, multiple equilibria, and path dependence.

3.1 What Utility Maximization Gets Wrong

The problem is not that utility maximization is false. Given a utility function and constraints, maximization is a reasonable model of choice. The problem is that the framework presupposes:

1. The utility function is known and stable
2. The constraints are exogenous
3. Agents optimize independently

When utilities are interdependent and context is endogenous, "maximize utility" doesn't define behavior—it merely redescribes whatever happens as "utility-maximizing."

The Identification Problem. If utility can include anything, any behavior is "rational" and the theory becomes unfalsifiable.

The Equilibrium Selection Problem. With interdependent utilities, multiple equilibria exist. Utility maximization says: at equilibrium, everyone is maximizing. It doesn't say which equilibrium will emerge.

The Stability Problem. Utility maximization describes optimal response to given conditions. It doesn't describe whether those conditions will persist.

4 A Framework with Complementarity and Context

4.1 From Optimization to Coherence

The limits of utility maximization suggest a different criterion: not "is each agent optimizing?" but "is the system of utilities coherent?"

Definition 2 (Coherence)**.** *A system is coherent if:*

- *Expectations match outcomes*
- *Institutions support behavior*
- *Norms align with incentives*
- *Identity is consistent with action*

4.2 The Extended Framework

Proposition 8 (Framework Extension)**.** *The required extension includes:*

1. *Allow $C_{ij} \neq 0$: Interdependent utilities with explicit complementarity parameter*

2. *Allow $\Psi_{t+1} \neq \Psi_t$: Endogenous context that evolves*

3. *Define K: Coherence as structural stability*

4. *Define Q: Quality as welfare at equilibrium*

5. *Analyze stability: When does $K \approx 1$ persist?*

The complementarity structure $C^*(\Psi)$ captures how utility interdependencies vary with context. This is not a minor extension—it transforms a single-agent optimization problem into a system problem.

5 Testable Predictions

The framework generates quantitative predictions about when behavioral deviations are large versus small.

Prediction 1 (Social Preference Magnitude)**.** *Social preference strength $\frac{\partial U_i}{\partial x_j}$ varies predictably:*

$$\frac{\partial U_i}{\partial x_j} = f(\mathit{distance}_{ij}, \mathit{merit}, \mathit{frame}, \mathit{stakes}) \tag{12}$$

where distance is social/psychological distance, merit captures perceived deservingness, frame captures cooperative vs. competitive framing, and stakes captures decision magnitude.

Hypothesis 1 (Context-Dependent Loss Aversion)**.** *Loss aversion λ varies with context:*

$$\lambda(\Psi) = f(\Psi_C, \Psi_T, \Psi_E, \mathit{domain}) \tag{13}$$

where Ψ_C is cognitive load, Ψ_T is temporal context, and Ψ_E is economic context.

Prediction 2 (Present Bias Heterogeneity)**.** *Present bias β varies across:*

- *Cognitive load: Higher load $\Rightarrow$ lower β (more present-biased)*

- *Stakes: Small stakes show larger present bias*
- *Domain: Health behaviors show more present bias than financial decisions*

Prediction 3 (Framing Effect Magnitude)**.** *Frame sensitivity varies inversely with preference crystallization:*

$$|\textit{Frame Effect}| \propto (1 - \textit{Preference Familiarity}) \tag{14}$$

Novel decisions show larger framing effects; familiar decisions show smaller effects.

6 Proposed Experimental Tests

6.1 Experiment 1: Context-Dependent Social Preferences

Design: $2 \times 2 \times 2$ factorial design varying:

1. Frame (cooperative vs. competitive)
2. Merit (earned vs. windfall endowment)
3. Social distance (in-group vs. out-group)

Outcome: Allocations in modified dictator game.

Prediction: Three-way interaction: social preferences highest in cooperative frame, earned endowment, in-group condition; lowest in competitive frame, windfall, out-group.

6.2 Experiment 2: Loss Aversion Under Cognitive Load

Design: Within-subject design measuring loss aversion under:

1. No load (control)
2. Working memory load (7-digit string)

3. Time pressure

Outcome: Accept/reject decisions for mixed gambles.

Prediction: Loss aversion increases under cognitive load: $\lambda_{load} > \lambda_{control}$.

6.3 Experiment 3: Present Bias and Commitment

Design: Longitudinal study with:

1. Baseline β elicitation
2. Commitment device availability manipulation
3. Follow-up behavior tracking

Prediction: Individuals with lower β (more present-biased) show higher demand for commitment devices and larger behavior improvement when devices are available.

6.4 Identification Strategy

Causal identification relies on:

1. Random assignment to context conditions
2. Within-subject designs where feasible
3. Pre-registered analysis plans
4. Multiple replications across populations

7 From Anomalies to Architecture

The behavioral economics literature spent decades documenting deviations from classical rationality. This was necessary and valuable—establishing *that* people deviate. But the next step is to predict *when* and *how much*.

Domain	What Varies	Calibration Target
Social preferences	Cultures, stakes, merit	$C_{ij}(\Psi_S, \Psi_I)$
Identity	Groups, priming, salience	$U^{IDN}(\Psi_S, \text{group})$
Reference dependence	Domains, expectations	$r(\Psi)$, $\lambda(\Psi)$
Framing	Presentations, defaults	Frame sensitivity $= f(\Psi_C)$
Preference construction	Familiarity, complexity	Construction weight $= f(\Psi_K)$
Temporal preferences	Cultures, cognitive load	$\beta(\Psi_C, \Psi_T)$

Table 2: Calibration opportunities from behavioral domains

Each experimental finding that documents context-dependence adds information to the functional form of $C^*(\Psi)$. The variation that makes behavioral economics "messy" is precisely what makes calibration possible.

8 Conclusion

The limits of utility maximization under stable preferences are not anomalies to be patched but structural features to be modeled. Six domains—social preferences, identity, reference dependence, framing, preference construction, and temporal discounting—share a common feature: utility interdependence generating non-zero complementarities.

The required extension does not reject utility maximization but embeds it within a richer structure where complementarity C and context Ψ are explicit parameters. This transforms the question from "what maximizes utility?" to "what pattern of interdependent utilities is coherent given context?"

The practical implications are substantial. Interventions that ignore context dependence will fail unpredictably. Policies designed for "representative agents" will miss the heterogeneity that determines aggregate outcomes. But the same features that make prediction difficult also make it possible: the variation across contexts is information, not noise.

Future work should calibrate context-dependent parameters from experimental variation and test the framework's predictions in field settings. The goal is not to abandon the optimization framework but to use it correctly—with the interdependencies and context

effects that decades of evidence have documented.

References

Almas, I., Cappelen, A. W., Sorensen, E. O., & Tungodden, B. (2020). Cutthroat capitalism versus cuddly socialism: Are Americans more meritocratic and efficiency-seeking than Scandinavians? *Journal of Political Economy*, 128(5), 1753–1788.

Andreoni, J. (1988). Why free ride? Strategies and learning in public goods experiments. *Journal of Public Economics*, 37(3), 291–304.

Ashraf, N., Karlan, D., & Yin, W. (2006). Tying Odysseus to the mast: Evidence from a commitment savings product in the Philippines. *Quarterly Journal of Economics*, 121(2), 635–672.

Berg, J., Dickhaut, J., & McCabe, K. (1995). Trust, reciprocity, and social history. *Games and Economic Behavior*, 10(1), 122–142.

Bryan, G., Karlan, D., & Nelson, S. (2010). Commitment devices. *Annual Review of Economics*, 2, 671–698.

Cappelen, A. W., Hole, A. D., Sorensen, E. O., & Tungodden, B. (2007). The pluralism of fairness ideals: An experimental approach. *American Economic Review*, 97(3), 818–827.

Cohen, J., Ericson, K. M., Laibson, D., & White, J. M. (2020). Measuring time preferences. *Journal of Economic Literature*, 58(2), 299–347.

Dana, J., Weber, R. A., & Kuang, J. X. (2007). Exploiting moral wiggle room: Experiments demonstrating an illusory preference for fairness. *Economic Theory*, 33(1), 67–80.

Fehr, E., & Charness, G. (2024). Social preferences: Fundamental characteristics and economic consequences. In *Handbook of Experimental Economics* (Vol. 3). Princeton University Press.

Fehr, E., & Schmidt, K. M. (1999). A theory of fairness, competition, and cooperation. *Quarterly Journal of Economics*, 114(3), 817–868.

Forsythe, R., Horowitz, J. L., Savin, N. E., & Sefton, M. (1994). Fairness in simple bargaining experiments. *Games and Economic Behavior*, 6(3), 347–369.

Frederick, S., Loewenstein, G., & O'Donoghue, T. (2002). Time discounting and time preference: A critical review. *Journal of Economic Literature*, 40(2), 351–401.

Güth, W., Schmittberger, R., & Schwarze, B. (1982). An experimental analysis of ultimatum bargaining. *Journal of Economic Behavior & Organization*, 3(4), 367–388.

Johnson, E. J., & Goldstein, D. (2003). Do defaults save lives? *Science*, 302(5649), 1338–1339.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291.

Kahneman, D., Knetsch, J. L., & Thaler, R. H. (1990). Experimental tests of the endowment effect and the Coase theorem. *Journal of Political Economy*, 98(6), 1325–1348.

Kőszegi, B., & Rabin, M. (2006). A model of reference-dependent preferences. *Quarterly Journal of Economics*, 121(4), 1133–1165.

Laibson, D. (1997). Golden eggs and hyperbolic discounting. *Quarterly Journal of Economics*, 112(2), 443–478.

Stigler, G. J., & Becker, G. S. (1977). De gustibus non est disputandum. *American Economic Review*, 67(2), 76–90.

Tversky, A., & Kahneman, D. (1981). The framing of decisions and the psychology of choice. *Science*, 211(4481), 453–458.

A 8D Document Type Specification

This paper was generated using the 8D Document Type Framework (Appendix DDD/CCC of the Evidence-Based Framework).

Style Profile: Ernst Fehr Academic Style

8D Coordinates:

- D_1 (Knowledge): 0.9 — Expert audience (behavioral economists, decision theorists)
- D_2 (Proximity): 0.9 — Same field (behavioral economics)
- D_3 (Scope): 0.7 — Field-wide impact
- D_4 (Time): 0.8 — Full journal article reading time
- D_5 (Goal): G1 — Inform (synthesize evidence, develop framework)
- D_6 (Context): 1.0 — External/public (journal publication)
- D_7 (Emotion): 0.1 — Highly factual
- D_8 (Persistence): 0.95 — Archival

Derived Structure (DT-5/DT-6):

- Technical glossary: Not required (expert audience)
- Detailed subsections: Yes (D4 $>$ 0.6)
- Theory section: Yes (D5 = G1)
- Full references: Yes (D8 $>$ 0.8)

Derived Style (DT-7/DT-8):

- Formal register: Yes (D6 $>$ 0.7)
- Hedging required: Yes (D8 $>$ 0.8)

- Technical vocabulary: Yes (D1 > 0.7)
- Impersonal style: Yes (D7 < 0.3)

B Relationship to EBF

This working paper synthesizes evidence presented in Chapter 3 of the Evidence-Based Framework for Economic and Social Behavior (EBF). The EBF is an integrative framework organizing 50+ years of behavioral economics evidence under the unified concepts of complementarity (C) and context (Ψ).

For complete formal derivations showing how each behavioral phenomenon emerges as a limiting case, see Appendix A of the EBF documentation.